BENJAMIN D. SHAFFER

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EXPERIENCE

Graduate Student Research Assistant, University of Pennsylvania August 2023 - Present
Physics-Informed Machine Intelligence Laboratory (PIMILab)*, Scalable Autonomous Robotics Laboratory* (ScalAR Lab)*, Philadelphia, PA*

Research Engineer, Captain, Air Force Research Laboratory October 2019 - October 2023
Laser Modeling and Simulation Section, Kirtland Air Force Base, Albuquerque, NM

EDUCATION

University of Pennsylvania, Philadelphia, PA August 2023 - Present
Ph.D. Student, Mechanical Engineering and Applied Mechanics
Advised by M. Ani Hsieh, Nathaniel Trask

Worcester Polytechnic Institute, Worcester, MA August 2015 - May 2019
BS, *with distinction*, Robotics Engineering
Advised by Jie Fu

FELLOWSHIPS

[1] National Science Foundation Graduate Research Fellowship 2024- 2027
[2] Walter L. Conwell Fellowship, University of Pennsylvania,
Department of Mechanical Engineering and Applied Mechanics 2023

PUBLICATIONS

[1] **Shaffer, B. D.**, Kinch, B., Hsieh, M. A., and Trask, N. (2026). *A meshfree exterior calculus for generalizable and data-efficient learning of physics from point clouds*. **NeurIPS *spotlight***. May 2026.

[2] **Shaffer, B.D.**, Koohy, S., Kinch, B., Hsieh, M.A. and Trask, N. *Structure-Preserving Learning Improves Geometry Generalization in Neural PDEs*. International Conference on Machine Learning (**ICML**). July 2026.

[3] **Shaffer, B.D.**, Edwards, V., Kinch, B., Trask, N. and Hsieh, M.A. *Multi-robot Multi-source Localization in Complex Flows with Physics-Preserving Environment Models*. International Conference on Robotics and Automation (**ICRA**). June 2026.

[4] **Shaffer, B.D.**, Vorenberg, J.R., and Hsieh, M.A. *Spectrally Informed Learning of Fluid Flows*. Chaos: An Interdisciplinary Journal of Nonlinear Science, 35(3), March 2024.

[5] **Shaffer, B.D.**, Vorenberg, J.R., Wilcox, C.C. and McDaniel, A.J., *Generalizable turbulent flow forecasting for adaptive optics control*. Applied Optics, 62(23), pp.G1-G11, August 2023.

[6] **Shaffer, B.D.**, McDaniel, A.J., Wilcox, C.C. and Ahn, E.S., *Dynamic mode decomposition based predictive model performance on supersonic and transonic aero-optical wavefront measurements*. Applied Optics, 60(25), pp.G170-G180, July 2021.

PREPRINTS

[1] **Shaffer, B.D.**, Kinch, B., Klobusicky, J., Hsieh, M.A. and Trask, N. *Physics-Informed Sensor Coverage Through Structure Preserving Machine Learning*. arXiv preprint arXiv:2509.10363. September 2025.

[2] Kinch, B., **Shaffer, B.D.**, Armstrong, E., Meehan, M., Hewson, J. and Trask, N. *Structure-Preserving Digital Twins via Conditional Neural Whitney Forms*. arXiv preprint arXiv:2508.06981. August 2025.

OTHER PROCEEDINGS

[1] Sahba, S., Wilcox, C. C., McDaniel, A., **Shaffer, B. D.** and Kutz, J. N. *Shallow Recurrent Decoder for Compressed Aero-Optical Wavefront Sensing*. In *Laser Science* (pp. JW4A-64). Optica Publishing Group, October 2023.

[2] **Shaffer, B.D.**, Vorenberg, J.R., Wilcox, C.C. and McDaniel, A.J., *Neural Network forecasting of transonic turbulent flow for adaptive optics control*. In Unconventional Imaging and Adaptive Optics. SPIE Optics + Photonics, August 2022.

[3] Sahba, S., Wilcox, C.C., McDaniel, A.J., **Shaffer, B.D.**, Brunton, S.L. and Kutz, J.N., *Wavefront sensor fusion for aero-optics via a shallow decoder neural network*. In Interferometry XXI. SPIE Optics + Photonics, August 2022.

[4] **Shaffer, B.D.**, McDaniel, A.J. and Wilcox, C.C., *Predictive modeling of wavefront error using dynamic mode decomposition*. In Interferometry XX (Vol. 11490, pp. 71-81). SPIE Optics + Photonics, August 2020.

INVITED TALKS

[1] **Shaffer, B.D.** "Geometric and Physics-Structured Learning for Robotics," General Robotics Actuation and Perception Lab, Student-Faculty-Industry Seminar. Available at: Structured learning for Robotics. September 2026.

[2] **Shaffer, B.D.** "Structure-Preserving Learning for Closed-Loop Digital Twins and Geometry Foundation Models," Department of Energy, Scalable, Efficient and Accelerated Causal Reasoning Operators, Graphs and Spikes for Earth and Embedded Systems (SEA-CROGS) Seminar. Available at: Structure-Preserving Learning. November 2025.

CONFERENCE PRESENTATIONS

[1] **Shaffer, B.D.** "Structure-Preserving Data-Driven Finite Element Models for General Geometries" WCCM / ECCOMAS joint meeting. July 2026.

[2] **Shaffer, B.D.**, "Data-driven Whitney forms for Structure Preserving Models", SIAM NNP Annual Meeting, November 2025

[3] Vorenberg, J.R., **Shaffer, B.D.**, "Transfer Learning Methods for Aero-Optical Flow Forecasting," 25th Directed Energy Professionals Society Annual Science and Technology Symposium, April 2023.

[4] **Shaffer, B.D.**, "Numerical Analysis of Transonic Flow over Hemispherical Turrets," AIAA Aviation, ITAR. June 2022.

[5] **Shaffer, B.D.**, "Variable Elevation Aero-Optical Effects Analysis for an Airborne Laser System," 24th Directed Energy Professionals Society Annual Science and Technology Symposium, April 2022.

[6] **Shaffer, B.D.**, "Tracking Algorithm Comparison for Aim-Point Maintenance in Directed Energy Applications," 24th Directed Energy Professionals Society Annual Science and Technology Symposium, April 2022.

[7] **Shaffer, B.D.**, "Numerical Analysis of Laser Pod Design" 24th Directed Energy Professionals Society Annual Science and Technology Symposium, April 2022.

[8] **Shaffer, B.D.**, "Beam Control Analysis for an Airborne Laser System," 23rd Directed Energy Professionals Society Annual Science and Technology Symposium, March 2021.

[9] **Shaffer, B.D.**, "HPC-enabled Digital Transformation for Laser Weapon Systems Development," DoD High Performance Computing Modernization Program Frontier Project Annual Review, April 2022.

[10] **Shaffer, B.D.**, "Data Driven Modeling of High Speed Flow," Directed Energy Joint Transition Office, Artificial Intelligence and Machine Learning Symposium, June 2021.

HONORS AND AWARDS

[1]	Air and Space Force Commendation Medal	2023
[2]	AFRL Laser Division Outstanding Publication	2020, 2021, 2022
[3]	AFRL Laser Division R. Earl Good Team Award	2021
[4]	United States Air Force Reserve Officer Training Corp Academic Scholarship	2015

PROFESSIONAL SERVICE

[1]	**Mentor, AFRL STEM Outreach**	October 2019 - August 2023
[2]	**Mentor, AFRL Scholars Internship Program** [x3]	May 2022 - August 2023
[3]	**Mentor, Big Brothers Big Sisters of Central New Mexico**	August 2021 - August 2023
[4]	**Reviewer,** NeurIPS, ICML, AAAI, ICRA, ICLR FM4Science, NPJ Climate and Atmospheric Science, Optica Applied Optics, Optica Optics Express, Neural Networks	